

ULIP Integration Requirement Document

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Revision History

Version	Date	Author	Reviewer	Significant Changes
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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for ACMES system information from ULIP system.

This document details the following points:

1. **Data Integration between ACMES and ULIP web service:** Identification of operational points where data integration between ACMES and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

User will share the information such as **AWB number**, **HAWB number** based on data send by user ULIP system will call respective **ACMES** API for accessing information.

Following are the point for accessing cargo information: -

1. This API will share data based on information provided by user.
2. ULIP will expose API for getting data from ACMES API.
3. **ACMES/01** API will take AWB number from user and get data from ACMES API.
4. **ACMES/02** API will take AWB number and HAWB number from user and get data from ACMES API.

1.2 Data Integration between ACMES System and ULIP application

User will share information to ULIP system based on data shared ULIP will provide data to user.

1.3 ACMES/01

This API will take **AWB number** and connect with ACMES API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	awbnumber	Master Airway bill Number as available in ACMES system.	[0-9]{11}	Required	Max 11

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **AWB number** for which detail is required.
3. ACMES API will share details of cargo information which exist in ACMES system in JSON format.
4. ACMES Integration will add JSON in **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

Example:

AWB number 40689348000 is shared by ULIP system, then ACMES-integration will make a request to ACMES API.

Request:

The request must be of the following format, where AWB number is passed within the URL body in JSON format.

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ACMES/01>

```
{  
  "awbnumber": "40689348000"  
}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ACMES/01' \  
--header 'authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzI3MTcwMzk5LCJhcHBzIjoZ  
GF0YXB1c2gifQ.djxBNGSkC8c4Tzkml9NfOML-O40nG9ucN6Qj2QVgqxnY_j-  
c0A2asTehZo3lArGRtxCYECdrGpQduZwSaWiy4w' \  
--header 'accept: application/json' \  
--header 'content-type: application/json' \  
--data '{  
  "awbnumber": "40689348000"  
}'
```

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ACMES/01>

```
{
  "awbnumber": "40689348000"
}
```

Curl URL

```
curl --location --request POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ACMES/01' \
--header 'authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsIm1hdCI6MTYzMDQ5MTk0Miwi
YXBwcyI6ImRhdGFwdXNoIn0.1NisXd9_2Vp7WpRCyqix7k6zvn9YuQS56PCZ6lvU
r8TRfrcbfprMdBeZT8vsWFD7ZYKsXmpf876N74uPgELhA' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data-raw '{
  "awbnumber": "40689348000"
}'
```

For obtaining authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of invalid format

1. Invalid AWB number

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at:
awbnumber. Format should follow [0-9]{11}"
}
```

b) In case of AWB number does not exist in ACMES system

```
{
  "response": [
    {
      "response": {
        "ExportTrackingData": {
          "AWB": null,
          "SHIPPING_BILL": [],

```

```

        "LOCATION_DETAILS": [],
        "XRAY_DETAILS": [],
        "ULD_DETAILS": [],
        "EGM_DETAILS": []
      },
      "responseStatus": "SUCCESS"
    },
    ],
    "error": "false",
    "code": "200",
    "message": "Success"
  }
}

```

c) In case AWB number exist

```

{"response":[{"response":{"ExportTrackingData":{"AWB":{"Destination":"DXB","P
ackages":"24","GrossWeight":"464","ChargeableWeight":"464","TDGDateTime":"08/
03/2021 16:57:00","BAGDateTime":"08/03/2021
21:13:00","GatePassDateTime":"N/A","HandOverDateTime":""},"SHIPPING_BILL":[],
"LOCATION_DETAILS":[],"XRAY_DETAILS":[{"XRayDate":"08/03/2021
20:33","XRayMachine":"40011512","XRayBy":"XRAY05"}],"ULD_DETAILS":[{"ULDNo":"
BT07IND","ULDType":"BT","Status":"A","FlightNo":"6E9602","FlightDate":"10/03
/2021"}],"EGM_DETAILS":[{"EGMNo":"341511","EGMYear":"11/03/2021","FlightNo":"
6E23","FlightDate":"11/03/2021","ATD":"","Commodity":"CONSOL
COURIER","Packages":"24","GrossWeight":"464"}]}}, {"responseStatus":"SUCCESS"}]
,"error":"false","code":"200","message":"Success"}

```

Status Code 200: cargo data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.4 ACMES/02

This API will take **AWB number** and **HAWB number** connect with ACMES API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	awbnumber	Master Airway bill Number as available in ACMES system.	[0-9]{11}	Required	Max 11

2	hawb	House Airway bill Number as available in ACMES system.	^[a-zA-Z0-9]{0,20}\$	Not-Required	Max 20
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1.4.1 Technical Approach

6. All the data will be shared through rest web services.
7. ULIP system will provide **AWB number and HAWB number** for which detail is required.
8. ACMES API will share details of cargo information which exist in ACMES system in JSON format.
9. ACMES Integration will add JSON in **response** key in response body.
10. ULIP system will provide the data in JSON Object format in response body.

Example:

AWB number **23552194682** and HAWB number **MUC75007115** is shared by ULIP system, then ACMES-integration will make a request to ACMES API.

Request:

The request must be of the following format, where AWB number and HAWB number is passed within the URL body in JSON format.

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ACMES/02>

```
{
  "awbnumber": "23552194682",
  "hawb": "MUC75007115"
}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ACMES/02' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTcwMzk5LCJhcHBzIjoiaZGF0YXB1c2gifQ.djxBNGSkC8c4Tzkm19NfOML-040nG9ucN6Qj2QVgqxnY_j-c0A2asTehZo3IARGrTxCYECdrGpQduZwSaWiy4w' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data '{
  "awbnumber": "23552194682",
  "hawb": "MUC75007115"
}'
```

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ACMES/02>

```
{
  "awbnumber": "23552194682",
  "hawb": "MUC75007115"
}
```

Curl URL

```
curl --location --request POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ACMES/02' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImhhdCI6MTYzMDU1MzM5NCwiYXBwcyI6ImRh
dGFwdXNoIn0.GJP8YAJ5hpyMP7S0tgc5PpVz4tQyu3NUkTifrDwD5WWDF0Q0LSrua5oi6tfY04yCd
MRiDVERD2nTn6fT5ByQog' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data-raw '{
  "awbnumber": "23552194682",
  "hawb": "MUC75007115"
}'
```

For obtaining authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of invalid format**1. Invalid AWB number**

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered
at: awbnumber. Format should follow [0-9]{11}"
}
```

2. Invalid HAWB number

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered
at: hawb. Format should follow ^[a-zA-Z0-9]{0,20}$"
}
```

b) In case of AWB number does not exist in ACMES system

```
{
  "response": [
    {
      "response": {
        "ImportTrackingData": {
          "AWB_DETAILS": null,
          "IGM_DETAILS": [],
          "DO_DETAILS": [],
          "GP_DETAILS": [],
          "LOC_DETAILS": []
        }
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

c) In case HAWB number does not exist in ACMES system

```
{
  "response": [
    {
      "response": {
        "ImportTrackingData": {
          "AWB_DETAILS": null,
          "IGM_DETAILS": [],
          "DO_DETAILS": [],
          "GP_DETAILS": [],
          "LOC_DETAILS": []
        }
      }
    }
  ]
}
```

```

    },
    "responseStatus": "SUCCESS"
  },
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}

```

d) In case AWB number and HAWB number exist

```

{
  "response": [
    {
      "response": {
        "ImportTrackingData": {
          "AWB_DETAILS": {
            "IgmNo": "1896697",
            "IgmYear": "2021",
            "AwbNo": "23552194682",
            "HawbNo": "MUC75007115",
            "ArrivedPackages": "1",
            "ArrivedWeight": "70",
            "ChargeableWeight": "192",
            "DamagedPackages": "0"
          },
          "IGM_DETAILS": [
            {
              "IgmNo": "1896697",
              "IgmYear": "2021",
              "FlightNo": "TK6448",
              "FlightDate": "16/03/2021",
              "SegregationDate": "16/03/2021 15:30",
              "SegregationStatus": "DONE",
              "ATA": "16/03/2021 04:45"
            }
          ],
          "DO_DETAILS": [
            {
              "DOIssuedBy": "TURKISH AIRLINES",
              "DODate": "17/03/2021 12:02:21",
              "HawbNo": "0",
              "DOIssuedTo": "SUNRISE FREIGHT FORWARDERS PVT.
LTD."
            }
          ],
        }
      }
    ]
  }
}

```

```

    {
      "DOIssuedBy": "SUNRISE FREIGHT FORWARDERS PVT.
LTD.",
      "DODate": "17/03/2021 12:34:28",
      "HawbNo": "MUC75007115",
      "DOIssuedTo": "REGENT GARAGE (P) LTD."
    },
    {
      "DOIssuedBy": "SUNRISE FREIGHT FORWARDERS PVT.
LTD.",
      "DODate": "17/03/2021 12:35:01",
      "HawbNo": "MUC75007116",
      "DOIssuedTo": "ASHWANI AUTOMOBILES PVT LTD."
    }
  ],
  "GP_DETAILS": [
    {
      "GatePassNo": "NN202103232098",
      "DeliveryPackages": "1",
      "GeneratedOn": "23/03/2021 11:48:00",
      "DocStatus": "DELIVERED"
    }
  ],
  "LOC_DETAILS": []
},
"responseStatus": "SUCCESS"
},
],
"error": "false",
"code": "200",
"message": "Success"
}

```

Status Code 200: cargo data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.5 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in ACMES system through a data exchange URL that has following form:

For production –

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ACMES/01>

Body:

```
{
  "awbnumber": "40689348000"
}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Brief summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl -X POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login'
-H 'accept: application/json'
-H 'content-type: application/json'
-data '{"username": "xxxxxx", "password": "XXXXX@12345"} '
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJpbmRlcmlvbnQpYXQjE1ODU1NjYyMzcsImFwcHMhOiJkYXRhcHVzaCJ9.jb-23Dbkwpz-3ptsuuZofzH-ce7jd0xR5WLkFfBPO_Dzjop01dSx90amqsjsgUARYLUPtUDiMX07oHG4L2AfFQ
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

```
curl --location --request POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ACMES/01' \
--header 'authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImIhdCI6MTYzMDU1MzM5NCwiYXBwcyI6ImRh
```

```
dGFwdXNoIn0.GJp8YAj5hpyMP7S0tgc5PpVz4tQyu3NUkTifrDwD5WwDF0Q0LSrua5oi6tfY04yCd
MRiDVERD2nTn6fT5ByQog' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data-raw '{
    "awbnumber": "40689348000"
}'
```

For staging –

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"
}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTcwMzk5LCJhcHBzIjoizGF0
YXB1c2gifQ.djxBNGSkC8c4Tzkml9NfOML-040nG9ucN6Qj2QVgqxnY_j-
c0A2asTehZo3IARGrTxCYECdrGpQduZwSawiy4w
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ACMES/01>

```
curl --Location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ACMES/01' \
--header 'authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTcwMzk5LCJhcHBzIjoizGF0
YXB1c2gifQ.djxBNGSkC8c4Tzkml9NfOML-040nG9ucN6Qj2QVgqxnY_j-
c0A2asTehZo3IARGrTxCYECdrGpQduZwSawiy4w' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data '{
    "awbnumber": "40689348000"
}'
```